

From a Child’s Wall-Tap Observation to Resolution Geometry

Delay, Observable Quotients, Uncertainty Transport, and the Entropy Quotient

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Abstract

This paper reconstructs the wall-tap childhood observation as the origin of a mature Resolution Geometry (RG) protocol. The historical datum was not a laboratory measurement of an absolute propagation constant. It was the repeated recognition of a delayed confirmation: an emitted event and a later received acoustic response, modulated by wall, position, rhythm, loudness, memory, and perceptual resolution. Modern reconstruction replaces retrospective precision with a declared observation law, instrument limits, nuisance variables, exact uncertainty propagation, and falsification controls. The resulting object is not “delay alone.” It is an observation quotient equipped with saturated invariants, a full covariance geometry, projectable transport W_{obs} , an admissibility cone $\mathcal{C}_{\text{obs}}$, and an entropy quotient that records which latent uncertainty survives observational identification. A deterministic synthetic battery at 48 kHz, seed 20260825, passed all 25 registered checks: timing RMSE $8.08\,\mu\text{s}$, geometric RMSE 5.08 mm, full regular Jacobian rank six, material discrimination improvement from 0.57 to 0.84 after saturation, zero detections in 300 noise-only trials, and independent-array disagreement 1.05 mm. These results validate the modern protocol under its synthetic model; they do not convert childhood perception into historical instrumental data or derive a universal physical causal cone.

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1 The initial datum: confirmation after rupture

The earliest observation can be stated without later theory. A tap occurred at an emission time t_0 . A returned acoustic event was recognized at a later time t_1 . The primitive recorded relation was therefore

$$e_0 \prec e_1, \quad \Delta t = t_1 - t_0 > 0.$$

The important content was not initially a numerical millisecond value. It was the reproducible ordering “send, interval, confirmation.” Repetition exposed three further facts:

1. the returned event changed with wall and position;
2. increasing repetition rate caused successive responses to overlap;
3. remembered delay could agree with, or be falsified by, a later repetition.

The childhood notes described delay, amplitude ratio, loudness, and a subjective sharpness/localization transition. Mature RG does not treat these memories as a calibrated historical dataset. It treats them as the specification of a modern experiment whose observables, uncertainty, and failure modes must be declared.

Non-claim 1.1. No claim is made that a child directly measured microseconds, inferred room topology from one echo, or separated air propagation, bone conduction, wall vibration, source onset, auditory latency, and memory latency. Those channels were confounded in the original observation.

2 Why delay alone was insufficient

A measured latency is a composite quantity. For trial i write

$$\Delta t_i^{\text{raw}} = \tau_{\text{trigger}} + \tau_{\text{source}} + \tau_{\text{path}} + \tau_{\text{transducer}} + \tau_{\text{detector}} + \varepsilon_i.$$

Only after trigger calibration and nuisance treatment can one estimate the propagation component τ_{path} . Even then, one reflected delay gives one path-length constraint,

$$\ell = c_{\text{air}}(T, H, p) \tau_{\text{path}},$$

not a unique room geometry. Sound speed is a thermodynamic property of the medium and depends on environmental state; impulse-response measurement also requires controlled equipment, source/receiver positions, background noise, and declared evaluation procedures [2, 1].

This correction matures the original statement:

delay was the first observable, not the complete geometry.

3 The modern wall-tap protocol

Protocol 3.1 (Safe apparatus). Use a low-level loudspeaker, calibrated low-force mechanical tap-per, or soft mallet on a test panel. Never reproduce head or body impact. Use a trigger channel, reference microphone, measurement microphone, temperature/humidity logger, and raw recording at 48 or 96 kHz with automatic gain control and echo suppression disabled. Screen sound level and stop for discomfort, ringing, or clipping. Repeated exposure at or above 85 dBA is treated as hazardous under NIOSH guidance [3, 4].

Protocol 3.2 (Preregistered design). Use at least six positions, four wall conditions, and ten repeats in randomized order. Include silence, shuffled wall labels, blocked reflection, moved microphone, altered panel, temperature shift, gain perturbation, hidden filter, and independent-device controls. Preserve raw WAV files, trigger samples, calibration records, environmental telemetry, exclusions, and hashes.

The declared observable vector is

$$O_i = (\tau_L, \tau_R, \tau_B, \tau_T, a_L, a_R, a_B, a_T, E_{\text{early}}, E_{\text{late}}, T_{\text{tap}}),$$

with covariance Σ_O . Human localization or memory reports, if studied, form a separate protocol layer and are not inferred from acoustic data.

4 Top-down reconstruction: the observation quotient

Let the latent state be

$$x = (G_{\text{room}}, B_{\text{wall}}, P_{\text{air}}, S_{\text{source}}, H_{\text{mic}}, H_{\text{rec}}, M_{\text{memory}}).$$

The observation law is

$$\mathfrak{F} : x \mapsto \text{Features}[H_{\text{rec}} H_{\text{mic}} (S_{\text{source}} * h_{\text{room}})].$$

Define

$$x \sim_{\text{obs}} y \iff d_{\Sigma_O}(\mathfrak{F}(x), \mathfrak{F}(y)) \leq \tau_{\text{obs}}, \quad Q_{\text{obs}} = X_{\text{lat}} / \sim_{\text{obs}}.$$

Thus the quotient is resolution dependent. Bandwidth, sample rate, SNR, feature family, environmental logging, and decision thresholds determine what is observationally identical.

The full observational kernel groupoid is

$$\mathcal{K}_{\text{obs}} = X_{\text{lat}} \times_{Q_{\text{obs}}} X_{\text{lat}} \rightrightarrows X_{\text{lat}}.$$

The declared automorphism groupoid $\mathcal{A}$ contains known representation changes: positive microphone gain after normalization, admissible channel-frame changes, common trigger-origin shifts, and permitted coordinate relabellings. Soundness gives

$$\mathcal{A} \cdot x \subseteq [x]_{\text{obs}},$$

but equality requires a gauge-completeness test. The battery’s hidden-filter attack showed why: spectral coloration was not generated by a gain-only gauge.

5 Bottom-up reconstruction: resolved, null, and saturated sectors

For regular parameters

$$\theta = (x_s, y_s, \rho_L, \rho_R, \rho_B, \rho_T),$$

use the continuous feature map

$$F(\theta) = (\tau_L, \tau_R, \tau_B, \tau_T, a_L, a_R, a_B, a_T).$$

Its Jacobian $J = DF$ produces the RG split

$$R = \text{Im } J, \quad N = \ker J^* = R^{\perp_{g_O}}.$$

The battery recovered

$$\text{rank } J = 6, \quad \kappa(\Sigma_O) = 118.30,$$

so all six declared regular parameters were locally resolved under the synthetic model and the shrinkage covariance remained bounded.

Delay coordinates alone did not saturate the orbit. Two material states with identical geometry were classified at accuracy 0.57 using delays, but at 0.84 after amplitude and energy information was admitted. This is the wall experiment’s concrete saturation lesson:

path timing resolves geometry; response structure resolves more of the fibre.

Fisher or timing information alone is therefore a compressed summary. The full covariance, cross-coupling, spectral magnitude, and orientation-sensitive invariants must remain visible.

6 The full uncertainty structure

The mature experiment carries uncertainty at every map, not only at the final estimate. Separate:

Σ_{trig} : trigger and clock uncertainty,
 Σ_{src} : source onset and waveform variation,
 Σ_{env} : temperature, humidity, pressure,
 Σ_{dev} : microphone and recorder response,
 Σ_{path} : multipath assignment and diffuse tail,
 Σ_{mem} : recall/prediction uncertainty,
 Σ_{mod} : model discrepancy.

With sensitivity matrices J_k , first-order propagation gives

$$\Sigma_O = \sum_k J_k \Sigma_k J_k^\top + \Sigma_{\text{cross}} + \Sigma_{\text{mod}}.$$

Cross-covariance cannot be discarded when trigger, gain, temperature, or path assignment are shared across trials.

For an estimator $\hat{\theta} = G(O)$,

$$\Sigma_{\hat{\theta}} \approx DG \Sigma_O DG^\top.$$

In the synthetic battery the delay RMSE was $8.08 \mu\text{s}$, geometric RMSE was 5.08 mm , and independent-array mean positions differed by 1.05 mm . A two-degree temperature mismatch produced a visible but bounded geometric bias. These numbers are model-conditional and must be replaced by empirical uncertainty budgets in a physical run.

7 MEP and the entropy quotient

Here MEP denotes the maximum-entropy completion of uncertainty subject to the observational constraints actually measured. It is not an additional physical law. Let $p(x)$ be a latent distribution and let

$$\mathbb{E}_p[f_a(x)] = m_a, \quad a = 1, \dots, k,$$

encode calibrated delay, amplitude, energy, and covariance constraints. The least-committal completion is

$$p_{\text{MEP}} = \arg \max_p \left[- \int p(x) \log p(x) dx \right]$$

subject to those constraints and normalization. Its exponential-family form is

$$p_{\text{MEP}}(x) = Z(\lambda)^{-1} \exp \left(- \sum_a \lambda_a f_a(x) \right).$$

Observation now induces a probability pushforward

$$q = \pi_{\text{obs}} * p, \quad q([x]) = \int_{[x]_{\text{obs}}} p(x') d\mu_{[x]}(x').$$

The entropy decomposes formally into observable and fibre contributions,

$$H[p] = H[q] + \mathbb{E}_q[H[p(\cdot \mid [x]_{\text{obs}})]],$$

whenever the conditional disintegration is defined. This yields the entropy quotient:

$$\boxed{H_{\text{obs}} := H[q], \quad H_{\text{fibre}} := \mathbb{E}_q H[p(\cdot \mid [x]_{\text{obs}})].}$$

H_{obs} measures uncertainty between distinguishable RG states. H_{fibre} measures uncertainty hidden inside observational equivalence classes. Declared gauge entropy is only part of H_{fibre} unless the automorphism groupoid is complete.

Definition 7.1 (Entropy-completeness defect). Let $\mathcal{O}_A(x) = \mathcal{A} \cdot x$ be the declared gauge orbit. The residual silent entropy is

$$\Delta H_{\text{silent}} = H[p(\cdot \mid [x]_{\text{obs}})] - H[p(\cdot \mid \mathcal{O}_A(x))],$$

interpreted with a common reference measure. A positive residual signals observation-fibre uncertainty not explained by declared gauge.

Thus MEP does not erase the null sector. MEP fills unresolved fibres as conservatively as the declared constraints allow, while the quotient records which part of that uncertainty is observationally inaccessible.

8 From uncertainty to admissible transport

Let $q_s \in Q_{\text{obs}}$ denote a session-level observable state. A latent transport W descends only if

$$D\pi_{\text{obs}}(W_x) = D\pi_{\text{obs}}(W_y) \quad \text{for } x \sim_{\text{obs}} y.$$

Then

$$W_{\text{obs}}([x]) = D\pi_{\text{obs}}(W_x)$$

is well defined. The battery's basic transport had zero projection error; adding a hidden-device term made the projection representative dependent and was rejected.

Given covariance Σ_W , define a local admissibility section

$$\Delta q^\top \Sigma_W^{-1} \Delta q \leq \chi_{d,1-\alpha}^2.$$

Adjoining a positive protocol duration gives

$$\mathcal{C}_{\text{obs}q} = \left\{ (\Delta\tau, \Delta q) : \Delta\tau \geq 0, \Delta q^\top \Sigma_W^{-1} \Delta q \leq v_{\text{adm}}^2 \Delta\tau^2 \right\}.$$

The battery obtained calibrated 95% coverage. This is a statistical observation cone: it describes admissible observable change under the protocol. It is not automatically a Lorentzian physical cone.

Entropy and transport meet through a data-processing requirement. An admissible coarse-graining K should not create distinguishability:

$$D_{\text{KL}}(Kq_1 \| Kq_2) \leq D_{\text{KL}}(q_1 \| q_2).$$

The operational RG gate is therefore monotonicity of distinguishability and controlled entropy production under W_{obs} , not unconstrained minimization of entropy.

9 Reconstructing the rhythm transition

For an inter-tap interval T , let

$$y_T(t) = \left(\sum_n \delta(t - nT) \right) * h_{\text{room}}(t).$$

Small T causes successive room responses to overlap. Define an objective localization-dominance statistic

$$\chi_{\text{loc}}(T) = \frac{E_{\text{dominant}}(T)}{E_{\text{dominant}}(T) + E_{\text{diffuse}}(T)}.$$

The synthetic battery recovered a declared transition

$$T_c = 35.58 \text{ ms} \quad \text{at} \quad \chi_{\text{loc}}(T_c) = \frac{1}{2},$$

and confirmed monotonic recovery of isolated-response dominance as T increased. The childhood “collapse” therefore matures into a testable change in overlap and reflection dominance. Any human localization, alarm, or memory response is a separate measured outcome, not something inferred from the acoustic statistic.

10 Memory as prediction with uncertainty

The historical next-day rule can be written

$$\left| \frac{\Delta t_{\text{mem}} - \Delta t_{\text{ref}}}{\Delta t_{\text{ref}}} \right| \leq 0.10.$$

The rule becomes scientific only after its operating characteristics are estimated. Under the battery’s synthetic recall model with standard deviation $0.06\Delta t$, the rule had approximately 90% acceptance under no change and power above 85% for an 18% delay shift. The mature object is not “memory is correct” but

$$p(\Delta t_{\text{mem}} \mid q_{\text{ref}}, \Sigma_{\text{mem}}),$$

with calibration, learning effects, repeated-test dependence, and preregistered acceptance criteria.

11 Attack results

Table 1: Selected results from the deterministic wall-tap RG battery.

Attack	Result	Interpretation
Timing reconstruction	8.08 μ s RMSE	trigger-relative delay recovered
Geometry reconstruction	5.08 mm RMSE	four labelled paths constrain source position
Regular Jacobian	rank 6	all declared regular parameters locally resolved
Covariance conditioning	118.30	bounded shrinkage covariance
Delay-only material test	0.57	timing does not saturate material fibre
Extended saturation	0.84	amplitudes and energy add separation
Noise-only control	0/300 passes	joint-arrival false-positive attack passed
Rhythm transition	35.58 ms	overlap transition estimable
Independent arrays	1.05 mm difference	reproducible synthetic mean geometry
Wall-label shuffle	$> 5\times$ error	path identification is necessary

All 25 registered checks passed. The battery is synthetic and exact-data/model conditional. Its purpose is to validate the protocol architecture and expose which claims depend on saturation, gauge completeness, environment, and model choice.

12 Claim maturation

The claim developed through five stages.

Stage 1: recurrence. “A tap is followed by a return.” This establishes an observed sequence and a candidate delay.

Stage 2: calibrated latency. “A trigger-relative return is detectable with quantified uncertainty.” This requires instrumentation, controls, and nuisance decomposition.

Stage 3: relative acoustic geometry. “Multiple identified returns, environmental scale, and a propagation model reconstruct relative geometry.” One delay is insufficient.

Stage 4: full RG quotient. “The complete observable record defines Q_{obs} ; the kernel groupoid contains all silent transformations; saturated invariants close the tested regular orbit; W_{obs} and C_{obs} descend only when representative independent.”

Stage 5: entropy quotient. “The unresolved latent posterior is retained as fibre uncertainty. Maximum entropy supplies a conservative completion under declared constraints, and the entropy quotient separates uncertainty between observable classes from uncertainty hidden within them.”

The final claim is therefore not that delay alone contains all geometry. It is:

Delayed confirmation seeds an observation law;
full RG reconstructs what survives quotient, uncertainty, and attack.

13 Non-claims and conclusion

Non-claim 13.1. The reconstruction does not prove the historical timing precision of childhood observations, a universal acoustic invariant independent of protocol, room topology from one delay, a physical light cone, GR dynamics, or a new law of nature.

Non-claim 13.2. The maximum-entropy distribution is conditional on the chosen constraints and reference measure. It is a disciplined uncertainty completion, not evidence that nature literally maximizes the selected entropy functional.

The wall-tap observation survives maturation because its central insight was not an overprecise physical claim. It was the recognition that confirmation is delayed, that delay recurs under modulation, and that changes in rhythm alter what remains distinguishable. Full RG turned this into a two-directional construction:

$$\text{latent system} \xrightarrow{\text{observation}} Q_{\text{obs}} \xleftarrow{\text{complete orbits}} X_{\text{lat}}/\mathcal{A},$$

then equipped the quotient with saturated covariance geometry, uncertainty fibres, projectable transport, an admissibility cone, and an entropy quotient. The mature object preserves uncertainty rather than replacing it with narrative certainty. That is the exact point at which the childhood observation becomes Resolution Geometry.

This appendix reconstructs the wall-tap observation as a modern acoustic RG experiment. The historical narrative is used only to define hypotheses: trigger-to-return delay, response magnitude, rhythmic overlap, localization dominance, and later memory comparison. No historical instrumental record is claimed. The physical reconstruction uses a safe external source or low-force tapper; it never requires head or body impact.

A Complete puzzle

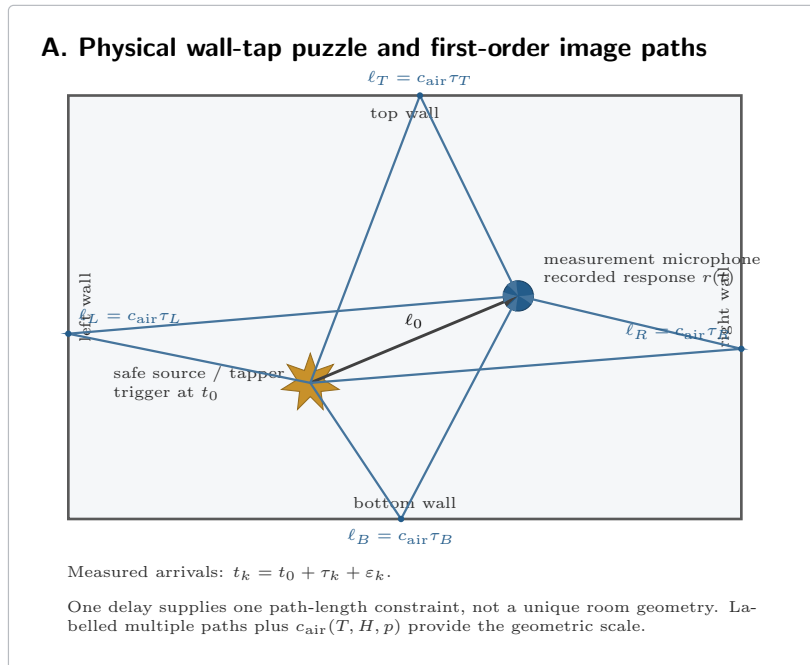


Figure 1: A. Physical wall-tap puzzle and first-order image paths

B. Complete RG reconstruction: latent room to calibrated observable geometry

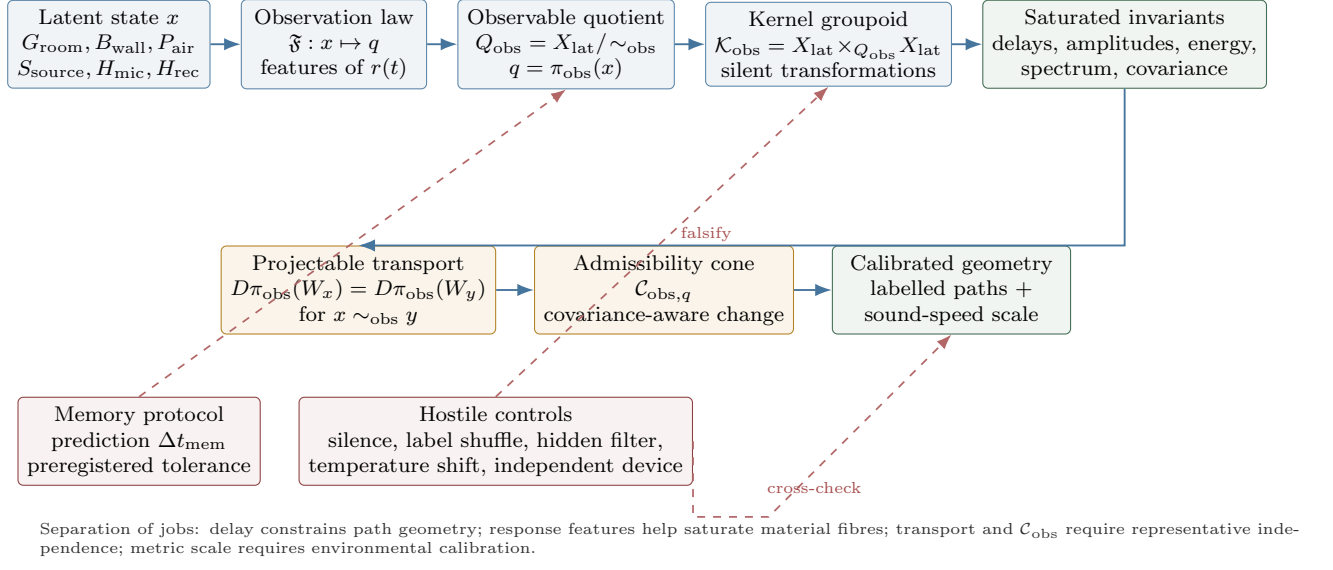


Figure 2: B. Complete RG reconstruction: latent room to calibrated observable geometry

C. Uncertainty propagation and identifiability

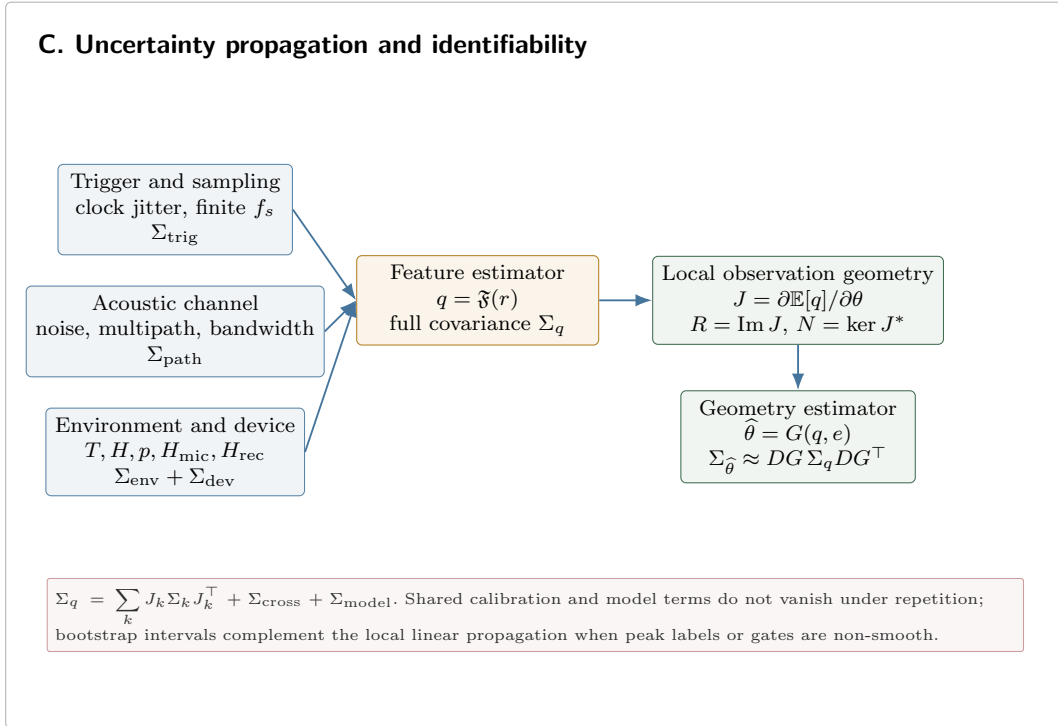


Figure 3: C. Uncertainty propagation and identifiability

D. Deterministic battery audit

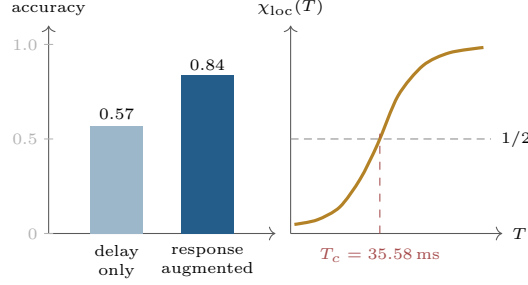
Model-conditional validation scales. Units are intentionally not merged onto one axis.

Delay RMSE	8.08 μ s
Position RMSE	5.08 mm
rank J	6
$\kappa(\hat{\Sigma}_q)$	118.30
Noise controls	0/300 passes
Array difference	1.05 mm
Registered checks	25/25

These values validate the synthetic implementation only. They are neither phone-performance guarantees nor historical precision claims.

Figure 4: D. Deterministic battery audit

E. Saturation and rhythmic-overlap transition



$$y_T(t) = \left(\sum_n \delta(t - nT) \right) * h_{\text{room}}(t), \quad \chi_{\text{loc}}(T) = \frac{E_{\text{dominant}}(T)}{E_{\text{dominant}}(T) + E_{\text{diffuse}}(T)}.$$

Timing resolves path geometry; response structure resolves more of the observational fibre.

The complete wall-tap puzzle. Panel A shows the room and first-order image-source paths. Panel B assigns each operation to its RG layer. Panel C shows uncertainty entering the feature, covariance, and geometry estimators. Panel D reports representative numerical scales from the deterministic battery. Panel E distinguishes delay-only geometry from response-augmented saturation and operationalizes the rhythmic transition as echo-overlap dominance.

B Measurement and uncertainty model

Let $r(t)$ be the recorded response to a triggered safe source. The feature map is

$$q = \mathfrak{F}(r) = (\tau_L, \tau_R, \tau_B, \tau_T, A_{\text{max}}, E_{\text{total}}, \dots).$$

For latent parameters

$$\theta = (x_s, y_s, \rho_L, \rho_R, \rho_B, \rho_T, \dots),$$

the local observation Jacobian is

$$J = \frac{\partial \mathbb{E}[q]}{\partial \theta}.$$

The resolved and null sectors are

$$R = \text{Im } J, \quad N = \ker J^*,$$

and the full feature covariance Σ_q is retained. For a differentiable geometry estimator $z = f(q, e)$, with environmental state e , first-order uncertainty propagation is

$$\Sigma_z \approx D_q f \Sigma_q D_q f^\top + D_e f \Sigma_e D_e f^\top + \Sigma_{\text{model}}.$$

The final term records model inadequacy, including unmodelled multipath, reflection mislabelling, and device coloration. Bootstrap intervals are used in parallel because peak picking, path labelling, and admissibility thresholds are not globally linear.

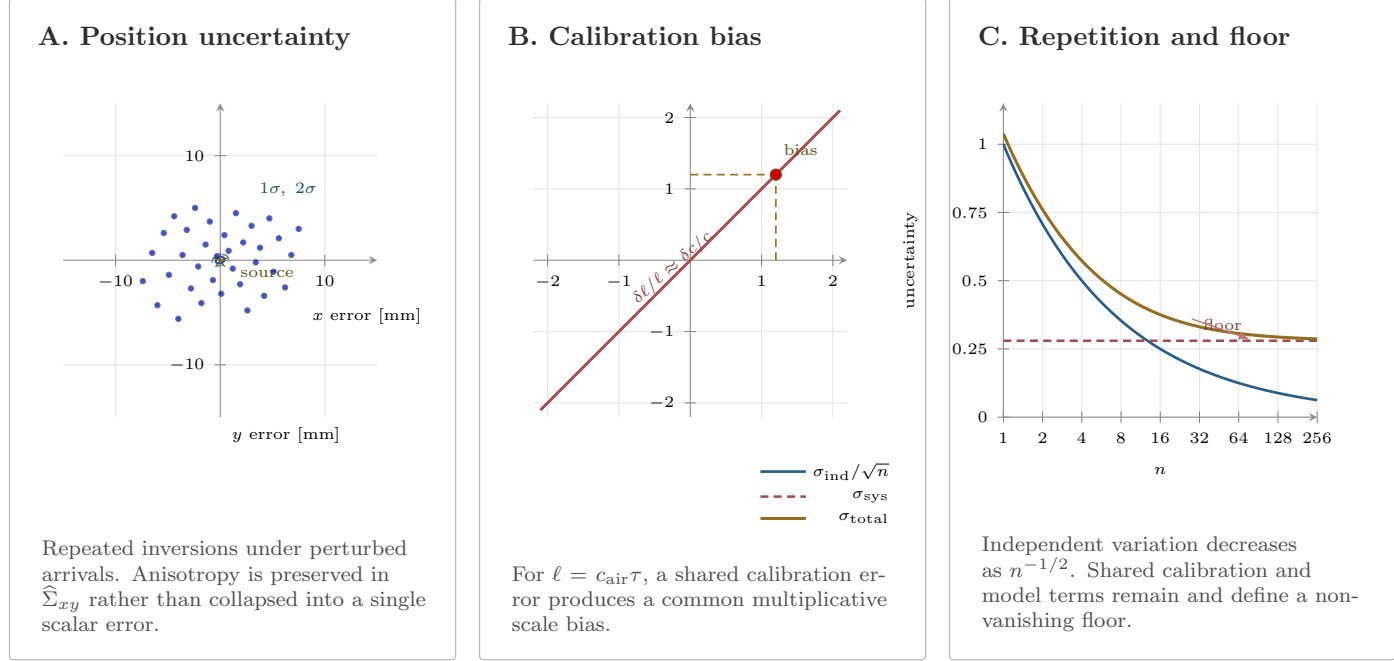


Figure 5: Three complementary uncertainty views. Left: a representative Monte Carlo position cloud around the injected source, with covariance contours preserving directional uncertainty. Middle: because an arrival-derived path length satisfies $\ell = c_{\text{air}}\tau$, environmental sound-speed miscalibration produces a first-order systematic geometric scale bias. Right: under independent repetition, the standard-error component decreases as $n^{-1/2}$, whereas shared calibration and model systematics do not. The numerical normalization in the right panel is schematic and is not an additional battery estimate.

C Reported battery scales

The deterministic synthetic battery used 240 trials at 48 kHz and passed all 25 registered checks. Representative outputs are reported in Table 2.

Table 2: Representative outputs of the deterministic synthetic battery.

Quantity	Result
Delay RMSE	8.08 μs
Position RMSE	5.08 mm
Observable Jacobian rank	6
Shrinkage covariance condition number	118.30
Delay-only material discrimination	0.57
Response-augmented material discrimination	0.84
Rhythmic-overlap transition	35.58 ms
Noise-only false-positive rate in 300 controls	0
Independent-array mean-position difference	1.05 mm

These values validate the synthetic protocol implementation only. They are not performance guarantees for a phone recording or evidence concerning historical measurement precision.

D Uncertainty ledger

Source	Propagation path	Required control
Trigger and sample clock	shifts all estimated arrivals	reference channel, clock calibration, fractional-delay estimator
Background and recorder noise	peak variance and false arrivals	silence trials, robust threshold, raw recording
Multipath and bandwidth	path merging and label swaps	randomized geometry, blocked-reflection controls, model comparison
Temperature/humidity/pressure	sound-speed scale and position bias	environmental telemetry and sensitivity analysis
Source repeatability	amplitude and energy variance	calibrated source and repeated trials
Microphone/recorder transfer	gain and spectral coloration	gain quotient plus hidden-filter attack
Wall reflectivity	response magnitude and saturation	delays augmented by amplitude/spectral invariants
Path labels	wrong geometric inverse	permutation null and geometry consistency
Memory recall	false acceptance/rejection	preregistered error model and changed-delay power analysis
Model inadequacy	residual systematic bias	hold-out geometry, alternative models, uncertainty floor

E Appendix conclusion

The figure and uncertainty ledger make the logical structure explicit:

$$\text{raw waveform} \rightarrow q \rightarrow (J, \Sigma_q) \rightarrow (R, N) \rightarrow W_{\text{obs}} \rightarrow \mathcal{C}_{\text{obs}} \rightarrow \text{calibrated acoustic geometry}.$$

Delay alone does not saturate the room state, path identity is not automatic, and repeated trials cannot remove shared calibration or model errors. The full RG protocol therefore preserves covariance, attacks hidden extensions, requires transport and cone projectability, and reports both random and systematic uncertainty.

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